Algebra Linear Equations & Slope Cheat Sheet

More Practice: writing equations of lines

1 FORMS OF A LINE

A linear equation can be written in different but equivalent forms. Each form is useful in different situations.

Slope-Intercept Form	Point-Slope Form	Standard Form
$y = mx + b$	$y - y_1 = m(x - x_1)$	$Ax + By = C$
m = slope b = y-intercept	m = slope (x_1, y_1) = a point on the line	A, B, and C are integers, and $A \geq 0$ $(A \neq 0)$

Example: Write the equation $y = 2x - 4$ in standard form.

Step 1: Move all terms to one side.

$\rightarrow 2x - y = 4$

Step 2: Write with A positive. $2x - y = 4$

Answer: $2x - y = 4$

2 WRITE AN EQUATION FROM SLOPE & A POINT

Use point-slope form, then simplify to slope-intercept form if needed.

Example: Write an equation of the line with slope $m = 3$ that passes through $(2, -1)$.

Step 1: Use point-slope form.

$y - (-1) = 3(x - 2)$

$y + 1 = 3(x - 2)$

Step 2: Distribute and simplify.

$y + 1 = 3x - 6$

$y = 3x - 7$

Answer: $y = 3x - 7$

You can leave your answer in point-slope form or simplify to slope-intercept form!

3 WRITE AN EQUATION FROM TWO POINTS

Find the slope using the two points, then use point-slope form (or slope-intercept form) to write the equation.

Example: Write the equation of the line through $(1, 2)$ and $(4, -1)$.

Step 1: Find the slope.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - 2}{4 - 1} = \frac{-3}{3} = -1$$

Step 2: Use one point, e.g., $(1, 2)$.

$y - 2 = -1(x - 1)$

$y - 2 = -x + 1$

$y = -x + 3$

Answer: $y = -x + 3$

4 FINDING INTERCEPTS

The x-intercept is where the line crosses the x-axis ($y = 0$).
The y-intercept is where the line crosses the y-axis ($x = 0$).

Example: Find the x- and y-intercepts of $2x - 3y = 12$.

x-intercept (set $y = 0$)

Step 1: $2x - 3(0) = 12$

$2x = 12$

$x = 6$

x-intercept: $(6, 0)$

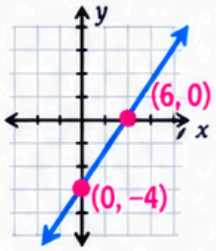
y-intercept (set $x = 0$)

Step 1: $2(0) - 3y = 12$

$-3y = 12$

$y = -4$

y-intercept: $(0, -4)$



5 PARALLEL AND PERPENDICULAR LINES

Parallel lines have the same slope.
Perpendicular lines have slopes that are negative reciprocals (the product of the slopes is -1).

Parallel Lines

Same slope: $m_1 = m_2$

Example: A line has slope 4. A parallel line also has slope 4.

Perpendicular Lines

Negative reciprocals: $m_1 \cdot m_2 = -1$ $\left(m_2 = \frac{1}{m_1}\right)$

Example: A line has slope 4. A perpendicular line has slope $-\frac{1}{4}$.

6 INTERPRET SLOPE AND INTERCEPTS

Slope and intercepts often have real-world meaning. Always include units in your interpretation.

Example: A bike rental costs \$5 plus \$3 per hour. Write the equation and identify the slope and y-intercept.

Let x = number of hours, and y = total cost (dollars).

$y = 3x + 5$

- **Slope: 3** → The cost increases by \$3 for each hour.
- **Y-intercept: 5** → The initial fee is \$5 (cost when $x = 0$).

The slope tells you the rate of change. The y-intercept tells you the starting value.

7 GRAPHING A LINE

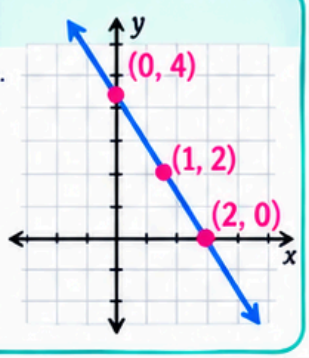
Use the slope and y-intercept to graph the line.

Example: Graph $y = -2x + 4$.

Step 1: Plot the y-intercept $(0, 4)$.

Step 2: Use the slope -2 (down 2, right 1) to find more points.

Step 3: Draw the line through the points.



8 QUICK REMINDERS & SPECIAL TIPS

DO REMEMBER:

- Slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Slope-intercept form: $y = mx + b$
- Point-slope form: $y - y_1 = m(x - x_1)$
- Parallel lines: same slope.
- Perpendicular lines: negative reciprocals.
- x-intercept: set $y = 0$.
- y-intercept: set $x = 0$.
- Write your final answer in the form requested.

COMMON MISTAKES:

- Reversing the slope formula.
- Forgetting to distribute in point-slope form.
- Confusing parallel and perpendicular slopes.
- Not solving for the correct intercept.
- Leaving fractions unsimplified.
- Forgetting to include units in real-world problems.

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Algebra Linear Equations & Slope Cheat Sheet

Unit Focus: *slope, rate of change, and writing linear equations*

1 KEY VOCABULARY

slope	The steepness of a line. The ratio of vertical change to horizontal change.	$m = 2$ (rises 2, runs 1)
rate of change	How much y changes for each 1 unit change in x.	y increases by 3 for every 1 increase in x. (rate = 3)
x-intercept	The point where a line crosses the x-axis. ($y = 0$)	(4, 0)
y-intercept	The point where a line crosses the y-axis. ($x = 0$)	(0, −3)
linear equation	An equation that graphs as a straight line.	$y = 2x - 5$
constant rate of change	The same change in y for every 1 unit change in x (a linear relationship).	y increases by 4 each time x increases by 1.

2 SLOPE AS RISE OVER RUN

$$\text{Slope } (m) = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x}$$

Example: Find the slope of the line through the points (1, 2) and (4, 6).

Step 1: Find the change in y (rise): $6 - 2 = 4$

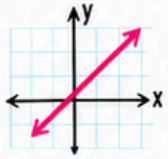
Step 2: Find the change in x (run): $4 - 1 = 3$

Step 3: Slope: $m = \frac{4}{3}$

Answer: $m = \frac{4}{3}$

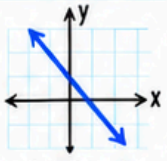
Types of Slope

Positive Slope ($m > 0$)



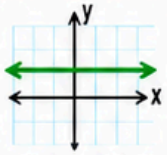
rises left to right ($m = 2$)

Negative Slope ($m < 0$)



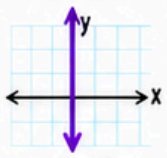
falls left to right ($m = -3$)

Zero Slope ($m = 0$)



horizontal line ($m = 0$)

Undefined Slope (no slope)



vertical line (undefined)

3 SLOPE FROM TWO POINTS

Use the slope formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Example: Find the slope through (−2, 3) and (4, −1).

Step 1: Substitute: $m = \frac{-1 - 3}{4 - (-2)}$

Step 2: Simplify: $m = \frac{-4}{6}$

Step 3: Simplify: $m = \frac{-2}{3}$

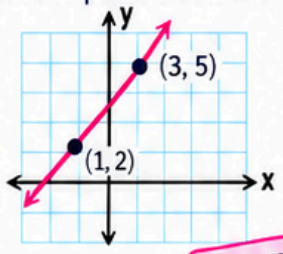
Answer: $m = \frac{-2}{3}$

Keep the order the same: $\frac{(y_2 - y_1)}{(x_2 - x_1)}$

4 FINDING SLOPE FROM A GRAPH OR TABLE

From a Graph

Pick two points on the line.



$m = \frac{5 - 2}{3 - 1} = \frac{3}{2}$

Answer: $m = \frac{3}{2}$

From a Table

Find the constant rate of change (slope) using any two rows.

x	y	
0	1	+3
1	4	+3
2	7	+3
3	10	

The y-value increases by 3 each time x increases by 1, so the slope is 3.

Answer: $m = 3$

5 SLOPE-INTERCEPT FORM

The slope-intercept form of a linear equation is $y = mx + b$,

m = slope b = y-intercept (the point where the line crosses the y-axis)

Example: Identify the slope and y-intercept of $y = -3x + 4$.

$m = -3$ (slope)

$b = 4$ (y-intercept)

Answer: slope = −3, y-intercept = 4, (0, 4)

The y-intercept is the point (0, b).

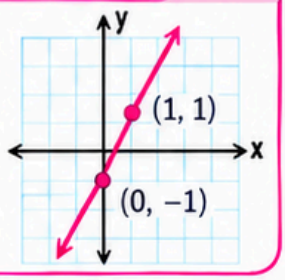
6 GRAPHING A LINE FROM SLOPE-INTERCEPT FORM

Graph the equation $y = 2x - 1$.

Step 1: Plot the y-intercept: (0, −1).

Step 2: Use the slope $m = 2$ (rise 2, run 1) to find another point.

Step 3: Draw a straight line through the points.



7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Slope is rise over run: $m = \frac{y_2 - y_1}{x_2 - x_1}$.
- A positive slope rises left to right; a negative slope falls left to right.
- The y-intercept is (0, b) in $y = mx + b$.
- A linear equation has a constant rate of change.
- Check your work by plotting points or using a table.

COMMON MISTAKES:

- Reversing the order in the slope formula. $\frac{(y_1 - y_2)}{x_1 - x_2}$ (*should be* $\frac{y_2 - y_1}{x_2 - x_1}$)
- Mixing up rise and run. *Using run/rise instead of rise/run.*
- Forgetting the sign when subtracting. $6 - 9 = -3$ (*not* 3)
- Thinking the y-intercept is the x-intercept. (0, b) (*not* (b, 0))
- Using a table with non-constant changes. *If the differences in y are not the same, it is not a linear relationship.*

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